

Reflection & Contribution Journal: Solo Submission

I am submitting this midterm project independently. Group coordination issues compromised the integrity of the initial submission, so I rebuilt this preprocessing pipeline from scratch. I executed 100% of the data engineering, documentation, and repository management. Typically, data leakage ruins predictive models. I prevented this immediately. I split the 7,043-row IBM Telco Churn dataset into a strict 70/30 train-test split before running a single transformation. The test set stayed in a vault.

Arguably, real-world data is dirty. The TotalCharges column contained hidden blank spaces masquerading as strings. I used regex coercion to force these into null values, then mathematically imputed them using the median of the training set. Useless identifiers like customerID were dropped entirely. Also, algorithms can't read text. I forced categorical variables into numeric matrices. Binary columns got strict Label Encoding. Nominal columns with multiple categories got One-Hot Encoded, and I aligned the test matrix to guarantee structural symmetry.

Critically, I didn't ignore the underlying mathematical distributions. Highly skewed numeric features were normalized using a Yeo-Johnson power transformation. I then squashed all continuous variables into a 0 to 1 range via MinMax scaling to prevent high-value metrics from overpowering the weights.

Significantly, the target variable was highly imbalanced. The model would have naturally been biased toward predicting "No Churn" simply because those instances outnumbered churners six-to-one. I deployed SMOTE (Synthetic Minority Over-sampling Technique) exclusively on the training set. This upsampled the minority class and forced a perfect 50/50 balance. Matplotlib visualizations in the notebook prove the transformation.

Individual Contribution Claim

I performed all dataset curation, project management, and Python coding. I claim full responsibility for the end-to-end preprocessing pipeline.